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BIOMEDICAL LIMITED

pH METER QUALIFICATION REPORT

Installation Qualification / Operational Qualification / Performance Qualification

LOCATION – WTP/Lab
PROTOCOL No. – BML/QA/pHQ/001
Effective date: August, 2025.
Overall Status:

PROVIDED BY:
BIOMEDICAL LIMTED
KWARA STATE, NIGERIA

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REPORT FORMAT APPROVAL:

Signatures on this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and date as shown below

	PREPARED BY:	**REVIEWED BY:**
Signature:		
Position:		
Date:		

	APPROVED BY:	**AUTHORIZED BY:**
Signature:		
Position:		
Date:		

1. Objective

This Qualification Report documents the results, conclusions, and approval status of the IQ, OQ, and PQ activities performed on the pH meter installed in the Plant at Biomedical Ltd (BML). The qualification was executed in accordance with Protocol BML/QA/pHQ/001 Version 01.

This report provides the formal documented evidence required by EU GMP Annex 15, USP <791>, and NAFDAC GMP Guidelines that the pH meter is fit for its intended purpose and approved for routine use in pharmaceutical manufacturing quality control.

2. Equipment Identification

Parameter	Details (to be completed at time of execution)
Equipment Name	pH Meter – Laboratory/Process Grade
Manufacturer	_____
Model	_____
Serial Number	_____
Electrode Type / Serial No.	_____
ATC Probe Serial No.	_____
Factory Calibration Cert. No.	_____
Location	QC Laboratory / WTP
Qualification Start Date	_____
Qualification End Date	_____

3. Pre-Qualification Checklist Results

Pre-Qualification Check	Criterion	Status	Init.
Purchase Order and delivery documentation available	Documents on file		
Instrument received undamaged – visual inspection	No damage		
Manufacturer's CoC / Factory Calibration Certificate received	CoC on file		
Approved operating SOP (BML/SOP/QC/005) in place	SOP current		
NIST-traceable buffer standards procured (pH 4.00, 7.00, 9.20)	Standards available		

Environmental conditions within specification	18–25°C / 30–70% RH		
Analyst trained on SOP	Training record		
Instrument registered in Calibration Register BML/QA/CR/001	Entry made		

Note: 'PENDING' cells to be updated to PASS/FAIL upon execution. All items must PASS before IQ commences.

4. Installation Qualification (IQ) Results

4.1 IQ Test 1 – Document Verification

Document	Acceptance Criterion	Reference / Finding	Status
URS	Approved URS on file prior to purchase		
Manufacturer's data sheet	Matches PO specifications		
Factory calibration certificate	NIST-traceable; within 12 months		
Operator's manual	Complete manual in English		
Packing list / invoice	All items received and accounted for		

4.2 IQ Test 2 – Physical and Hardware Verification

Check	Acceptance Criterion	Observed	Status
Main unit model	Matches PO and URS		
Serial number	Matches CoC / invoice		
Physical condition	No damage, cracks, or corrosion		
Display screen	Fully legible, no dead pixels		
Keypad / buttons	All functional; no sticking keys		
Electrode socket	Clean, undamaged		
ATC probe socket	Present and undamaged		
Power adaptor / cable	Correct voltage; cable undamaged		
BML asset tag affixed	BL/EQ/QAE/ visible		
Installation location	Level; away from vibration and direct sunlight		
Grounding	Properly earthed per manufacturer		

4.3 IQ Test 3 – Utilities Verification

Utility	Acceptance Criterion	Observed	Status
Mains voltage	Within manufacturer's range (e.g., 220–240 V AC ±10%)		
Power socket	Stable; UPS preferred		
Environmental temperature	18–25°C		
Relative humidity	30–70% RH		
Bench area	Non-metallic, level, with drainage proximity		

4.4 IQ Conclusion

All IQ tests are pre-populated as PENDING. Upon execution, each test outcome shall be recorded in the above tables and the following conclusion completed.

IQ Outcome	Details
Overall IQ Status	
Number of deviations	
Deviation references	
IQ Conclusion statement	*The pH meter has been / has not been [delete as applicable] successfully installed in accordance with Protocol BML/QA/pHQ/001.*

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

5. Operational Qualification (OQ) Results

Buffer standards used and traceability details to be recorded in the table below at time of execution.

Buffer	Nominal pH at 25°C	Lot / Expiry	Source	Traceability
Buffer 1	4.00 ± 0.01			NIST-traceable
Buffer 2	7.00 ± 0.01			NIST-traceable
Buffer 3	9.20 ± 0.01			NIST-traceable

5.1 OQ Test 1 – Power-On and Self-Diagnostic Results

Test	Acceptance Criterion	Result	Status
Power-on sequence	Powers on without error codes		
Self-diagnostic / RAM test	Self-test completes successfully		
Display initialisation	All segments visible; units displayed correctly		
Date and time setting	Clock accurate; retained after power cycle		
Electrode connection recognition	Recognised without error		
ATC probe response	ATC temperature = ambient ± 0.5°C		

5.2 OQ Test 2 – Single-Point Calibration Verification

Replicate	Buffer pH	Displayed pH	Deviation	Status
1	7.00			
2	7.00			
3	7.00			
Acceptance	±0.05 pH	Mean:		

5.3 OQ Test 3 – Two-Point Calibration and Linearity

Buffer	Reference pH	Displayed pH	Deviation	Status
pH 4.00 buffer	4.00			
pH 7.00 buffer	7.00			
pH 9.20 buffer	9.20			

Acceptance: ±0.05 pH units at each point.

5.4 OQ Test 4 – Electrode Slope Check

Parameter	Acceptance Criterion	Observed	Status
Electrode slope (%)	95.0 – 105.0%		
Asymmetry potential (mV)	≤ ±30 mV		

Note: A slope below 95% indicates electrode degradation. Replace electrode and re-test before proceeding.

5.5 OQ Test 5 – Temperature Compensation Verification

Buffer	Target Temp (°C)	Actual Temp (°C)	Expected pH	Displayed pH	Status
pH 7.00	20		7.00 ± 0.05		
pH 7.00	37		7.00 ± 0.05		

5.6 OQ Test 6 – Response Time

Test	From Buffer	Time to Stable Reading (s)	Criterion	Status
Response time	pH 7.00 → pH 4.00		≤ 60 s	
Response time (reverse)	pH 4.00 → pH 9.20		≤ 60 s	

5.7 OQ Test 7 – Repeatability (10 Consecutive Readings)

Reading	pH Displayed	Reading	pH Displayed	Criterion
1		6		
2		7		RSD ≤ 0.3%
3		8		
4		9		All ± 0.05 pH
5		10		
Mean ± SD		Mean: _____ SD: _____ RSD: _____%		Status:

5.8 OQ Conclusion

OQ Outcome	Details
Overall OQ Status	*PENDING – complete upon execution*
Number of deviations	
Deviation references	
OQ Conclusion statement	*The pH meter has been / has not been demonstrated to operate within specified limits as required by Protocol BML/QA/pHQ/001.*

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

6. Performance Qualification (PQ) Results

6.1 PQ Test 1 – Inter-Day Precision (3 Days)

Day	Analyst	Buffer pH	Displayed pH	Deviation	Criterion	Status
Day 1		7.00			±0.05 / RSD ≤0.5%	
Day 2		7.00			±0.05 / RSD ≤0.5%	
Day 3		7.00			±0.05 / RSD ≤0.5%	

6.2 PQ Test 2 – pH Measurement of LVP Products

All measurements to be performed in triplicate on representative samples from approved batches or reference samples. Results compared against USP/Ph.Eur. specifications.

Product	Specification (USP)	Result 1	Result 2	Result 3	Mean	Status
Normal Saline 0.9%	4.5 – 7.0					
Dextrose 5% (D5W)	3.5 – 6.5					
Ringer's Lactate	6.0 – 7.5					

6.3 PQ Test 3 – Accuracy Against External Reference

Sample	Reference pH (Certified)	BML Measured pH	Difference	Status
Sample A				
Sample B				

Acceptance: BML result within ±0.05 pH of certified reference.

6.4 PQ Test 4 – Operator-to-Operator Variability

Analyst A	pH Reading	Analyst B	pH Reading	Statistical Summary
Reading 1		Reading 1		Analyst A Mean:
Reading 2		Reading 2		Analyst B Mean:
Reading 3		Reading 3		Combined RSD:
Reading 4		Reading 4		Mean Difference:
Reading 5		Reading 5		Overall Status:

Acceptance: Combined RSD ≤ 0.5%; mean difference between analysts ≤ 0.05 pH.

6.5 PQ Conclusion

PQ Outcome	Details
Overall PQ Status	
Number of deviations	
Deviation references	
PQ Conclusion statement	*The pH meter has been / has not been demonstrated to perform consistently for its intended use in LVP pharmaceutical manufacturing.*

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

7. Deviation Log

All deviations observed during qualification activities shall be recorded in this log. No deviations identified = enter 'None'.

Dev. No.	Test Ref.	Date	Description of Deviation	CAPA / Corrective Action	Status
1			*None*	*N/A*	*Closed*

8. Overall Qualification Summary and Release Decision

Qualification Phase	Scope	# Tests	Overall Status
Pre-Qualification Checks	8 administrative and environmental checks	8	
Installation Qualification (IQ)	Document verification, physical checks, utilities	3 test groups	
Operational Qualification (OQ)	Power-on, calibration, linearity, slope, ATC, response time, repeatability	7 tests	
Performance Qualification (PQ)	Inter-day precision, product testing, external accuracy, operator variability	4 tests	

8.1 Final Qualification Conclusion

CONCLUSION (to be completed upon execution):

Based on the qualification activities performed in accordance with Protocol BML/QA/pHQ/001 Version 01, the pH meter (Equipment Tag:) has been / has not been demonstrated to be:

- Correctly installed in accordance with manufacturer's specifications and BML requirements (IQ PASSED / FAILED)
- Operating within defined performance limits across its intended measurement range (OQ PASSED / FAILED)

- Consistently performing accurate pH measurements suitable for LVP pharmaceutical manufacturing QC (PQ PASSED / FAILED)

The pH meter is therefore APPROVED / NOT APPROVED for routine use. Routine calibration shall be performed daily (before use) in accordance with BML/QDG/QA/ , with quarterly external calibration as per BML/QA/CAL/001.

8.2 Final Release Signatures

Role	Name	Signature	Date
QA officer			
QA Manager			
Factory Manager			

9. Post-Qualification Requirements and Routine Maintenance

Activity	Details	Frequency	SOP Ref.
Routine calibration (2-point)	pH 4.00 and pH 7.00 buffers; before each use	Daily / Per use	
Electrode conditioning	Store electrode in KCl solution; condition in pH 7.00 buffer before use	Daily	
Slope check	Verify slope 95–105% after 2-point calibration	Daily	
Electrode visual inspection	Check for cracks, contamination, or electrolyte depletion	Weekly	
External calibration	By accredited NIST-traceable calibration laboratory	Annual	
Requalification review	Abbreviated PQ using 3 LVP products; review calibration trends	Annual	This report
Electrode replacement	Replace when slope <95% or asymmetry >30 mV; trigger OQ+PQ	As needed	

10. Document Control

Version	Date	Description of Change	Approved by
01	April 2026	Initial issue – qualification report template	QA Manager

11. Appendix – Raw Data Attachment Sheet

The following raw data worksheets shall be attached to this report upon completion of qualification activities:

#	Attachment Description	Document Reference	Attached (Y/N)
A	Completed IQ data sheets with analyst initials and dates	Section 7	
B	Completed OQ data sheets (all 7 tests)	Section 8	
C	Completed PQ data sheets (all 4 tests)	Section 9	
D	Manufacturer's factory calibration certificate	CoC No.: _____	
E	Buffer standard certificates of analysis	Lot No.(s): _____	
F	Analyst training records for all personnel who executed tests	BML Training System	
G	Deviation forms (if any)	DEV/	
H	Environmental monitoring records at time of qualification	Temp/RH log	